

STAT 339 METHODS OF LINEAR ALGEBRA MATRICES

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October 10, 2023



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① Matrices

- Operations with Matrices

- Properties of Matrix Operations

- Exercise 1

- The Inverse of a Matrix

- Elementary Matrices

- Exercise 2

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Definition

Matrix

A **matrix** is a rectangular (or square) array of numbers or variables. E.g.

$$\mathbf{A} = [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} \in M_{m \times n}$$

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where

- a_{ij} is (i, j)-th entry (or element),
- m is number of rows,
- n is number of columns,
- $m \times n$ is the size, or **dimension**.

Definition

Square Matrix

A **Square Matrix** is simply a matrix with the same number of rows as number of columns. That is, $m = n$.

$$A = [a_{ij}]_{n \times n} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix}_{n \times n} \in M_{n \times n}$$

Definition

Vector

A **vector** is simply a matrix with only one column. E.g.

$$\mathbf{a} = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \\ \vdots \\ a_{n1} \end{bmatrix}_{n \times 1} \in M_{n \times 1},$$

Definition

Vector

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$$\mathbf{a} = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \\ \vdots \\ a_{n1} \end{bmatrix}_{n \times 1} \in M_{n \times 1},$$

is the vector formed from the first column of $\mathbf{A}$ above.

- Vectors will always be column vectors. If we need a row vector we will use the transpose of a vector. E.g., $\mathbf{a}^T = (a_{11}, a_{21}, a_{31}, \dots, a_{n1})$ is the row vector version of $\mathbf{a}$, where T denotes matrix transpose.

Definition

Partitioned matrices

$$A = [a_{ij}]_{3 \times 4} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}_{3 \times 4} = \begin{bmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \mathbf{r}_3 \end{bmatrix}_{3 \times 1}$$

$$A = [a_{ij}]_{3 \times 4} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}_{3 \times 4} = \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \mathbf{c}_3 & \mathbf{c}_4 \end{bmatrix}_{1 \times 4}$$

$$A = [a_{ij}]_{3 \times 4} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}_{3 \times 4} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}_{2 \times 2}$$

Definition

Scalar

A **Scalar** is a 1×1 matrix; i.e., a real-valued number or variable. Scalars will be denoted in ordinary (non-bold) typeface.

Diagonal matrix

A **diagonal matrix** is a square matrix with all of its off-diagonal elements equal to 0. We will use the `diag(.)` function in two ways: if its argument is a square matrix, then `diag(.)` yields a vector formed from the diagonal of that matrix; if its argument is a vector, then `diag(.)` yields a diagonal matrix with that vector on the diagonal. E.g.,

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$$\text{diag} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ 9 \end{pmatrix},$$

Definition

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$$\text{diag} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ 9 \end{pmatrix}, \quad \text{diag} \begin{pmatrix} 1 \\ 5 \\ 9 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 9 \end{pmatrix} \in M_{3 \times 3}$$

Definition

Identity matrix

The $n \times n$ **identity matrix** is a diagonal matrix with 1's along the diagonal. We will denote this as I , or I_n when we want to make clear what the dimension is. E.g.,

$$I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- The identity matrix has the property that

$$IA = A, \quad BI = B,$$

where A, B and I are assumed conformable to these multiplications.

Vector & matrix of 1's

A vector of 1's is denoted as $\mathbf{j}$, or $\mathbf{j}_n$ when we want to emphasize that the dimension is n . A matrix of 1's is denoted as $\mathbf{J}$, or $\mathbf{J}_{n,m}$ to emphasize the dimension. E.g.,

$$\mathbf{j}_4 = (1, 1, 1, 1)^T \quad \mathbf{J}_{2,3} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

- A vector or matrix containing all 0's will be denoted by $\mathbf{0}$. Sometimes we will add subscripts to identify the dimension of this quantity.

Definition

Lower & upper-triangular matrices

Lower and upper-triangular matrices have 0's above and below the diagonal, respectively. E.g.,

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 4 & 2 & 0 \\ 5 & 6 & 4 \end{pmatrix}, \quad U = \begin{pmatrix} 8 & 6 & 7 \\ 0 & 5 & 9 \\ 0 & 0 & 3 \end{pmatrix}.$$

Trace operation

If $A = [a_{ij}]_{n \times n}$, then $\text{Tr}(A) = \sum_{i=1}^n a_{ii}$

Definition

Equal matrices

Two matrices are equal if they have the same size ($m \times n$) and entries corresponding to the same position are equal. That is,

For $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{m \times n}$, $A = B$ if and only if $a_{ij} = b_{ij}$ for $1 \leq i \leq m$, $1 \leq j \leq n$,

Ex1: Equality of Matrices

Given

$$A = \begin{pmatrix} 1 & 7 & 0 \\ 4 & 2 & 9 \\ 5 & 6 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}.$$

If $A = B$, then $a = 1, b = 7, c = 0, d = 4, e = 2, f = 9, g = 5, h = 6$ and $i = 4$.

Sums and Scalar Multiples

If A and B are $m \times n$ matrices, r a scalar,

- **Sum:** $A + B = [a_{ij}]_{m \times n} + [b_{ij}]_{m \times n} = [a_{ij} + b_{ij}]_{m \times n} = [c_{ij}]_{m \times n} = C$

Linear Operations

Sums and Scalar Multiples

If A and B are $m \times n$ matrices, r a scalar,

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- **Scalar multiplication:** $rA = r[a_{ij}]_{m \times n} = [ra_{ij}]_{m \times n}$

Linear Operations

Sums and Scalar Multiples

If A and B are $m \times n$ matrices, r a scalar,

- **Sum:** $A + B = [a_{ij}]_{m \times n} + [b_{ij}]_{m \times n} = [a_{ij} + b_{ij}]_{m \times n} = [c_{ij}]_{m \times n} = C$
- **Scalar multiplication:** $rA = r[a_{ij}]_{m \times n} = [ra_{ij}]_{m \times n}$
- **Subtraction:** $A - B = A + (-1)B$

Ex2: Matrix addition, Scalar multiplication and matrix subtraction

Given

$$A + B = \begin{pmatrix} 1 & 7 \\ 4 & 2 \end{pmatrix} + \begin{pmatrix} 3 & 5 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1+3 & 7+5 \\ 4+1 & 2+0 \end{pmatrix} = \begin{pmatrix} 4 & 12 \\ 5 & 2 \end{pmatrix}.$$

$$3A - B = 3 \begin{pmatrix} 1 & 7 \\ 4 & 2 \end{pmatrix} + (-1) \begin{pmatrix} 3 & 5 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 3 & 21 \\ 12 & 6 \end{pmatrix} + \begin{pmatrix} -3 & -5 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 3-3 & 21-5 \\ 12-1 & 6-0 \end{pmatrix} = \begin{pmatrix} 0 & 16 \\ 11 & 6 \end{pmatrix}$$

Definition

Matrix multiplication

If $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{n \times p}$, then

$$AB = [a_{ij}]_{m \times n} [b_{ij}]_{n \times p} = [c_{ij}]_{m \times p} = C,$$

where

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj} = a_{i1} b_{1j} + a_{i2} b_{2j} + \cdots + a_{in} b_{nj}.$$

NOTE: The entry c_{ij} is obtained by calculating the sum of the entry-by-entry product between the i -th row of A and the j -th column of B .

Definition

Matrix multiplication

If $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{n \times p}$, then the product AB is $m \times p$ matrix whose columns are $Ab_1, \dots, Ab_p$. That is,

$$AB = A[b_1 \quad b_2 \quad \dots \quad b_p] = [Ab_1 \quad Ab_2 \quad \dots \quad Ab_p].$$

NOTES

- 1 Multiplication of matrices corresponds to composition of linear transformations.

Matrix multiplication

Ex3: Find AB , if

$$A = \begin{pmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{pmatrix}_{3 \times 2}, \quad B = \begin{pmatrix} -3 & 2 \\ -4 & 1 \end{pmatrix}_{2 \times 2}$$

$$AB = \begin{pmatrix} (-1)(-3) + (3)(-4) & (-1)(2) + (3)(1) \\ (4)(-3) + (-2)(-4) & (4)(2) + (-2)(1) \\ (5)(-3) + (0)(-4) & (5)(2) + (0)(1) \end{pmatrix}_{3 \times 2} = \begin{pmatrix} -9 & 1 \\ -4 & 6 \\ -15 & 10 \end{pmatrix}_{3 \times 2}$$

Matrix multiplication

$$Ab_1 = \begin{pmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{pmatrix} \begin{pmatrix} -3 \\ -4 \end{pmatrix} = \begin{pmatrix} -9 \\ -4 \\ -15 \end{pmatrix} \quad Ab_2 = \begin{pmatrix} -1 & 3 \\ 4 & -2 \\ 5 & 0 \end{pmatrix} \begin{pmatrix} -3 \\ -4 \end{pmatrix} = \begin{pmatrix} 1 \\ 6 \\ 10 \end{pmatrix}$$

Then

$$AB = A[b_1 \quad b_2] = \begin{pmatrix} -9 & 1 \\ -4 & 6 \\ -15 & 10 \end{pmatrix}_{3 \times 2}$$

NOTES

- 1 BA is not multipliable,
- 2 Even BA is multipliable, it could be that $AB \neq BA$.

Matrix form of a system of linear equations in n variables

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$



$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$\parallel$
 A

$\parallel$
 $\mathbf{x}$

$\parallel$
 $\mathbf{b}$

m linear equations



single matrix equation

$$\underset{m \times n}{A} \underset{n \times 1}{\mathbf{x}} = \underset{m \times 1}{\mathbf{b}}$$

Application

$A\mathbf{x}$ is a linear combination of the column vectors of matrix A

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} = [\mathbf{c}_1 \quad \mathbf{c}_2 \quad \cdots \quad \mathbf{c}_n] \text{ and } \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{bmatrix}$$

$$\begin{aligned} A\mathbf{x} &= \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_m \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_m \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_m \end{bmatrix}_{m \times 1} = x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} \\ &= x_1 \mathbf{c}_1 + x_2 \mathbf{c}_2 + \cdots + x_m \mathbf{c}_n \end{aligned}$$

Properties of matrix addition and scalar multiplication

Theorem

Let A , B and C be matrices of the same size, and let r and s be scalars, then

- ① $A + B = B + A$, (Commutative property of matrix addition)
- ② $(A + B) + C = A + (B + C)$, (Associative property of matrix addition)
- ③ $1A = A$, (Multiplicative identity property, and 1 is the multiplicative identity for all matrices)
- ④ $r(A + B) = rA + rB$, (Distributive property of scalar multiplication over matrix addition)
- ⑤ $(r + s)A = rA + sA$, (Distributive property of scalar multiplication over real-number addition)
- ⑥ $r(sA) = (rs)A$, (Associative property of scalar multiplication)

NOTES

- All above properties are very similar to the counterpart properties for real numbers

Properties of zero matrices

Theorem

Let A be matrices of the size $m \times n$, and let r be scalar, then

- ① $A + \mathbf{0}_{m \times n} = A$,
(Additive identity property, and $\mathbf{0}_{m \times n}$ is the Additive identity for all matrices)
- ② $A + (-A) = \mathbf{0}_{m \times n}$,
(Thus, $-A$ is called the additive inverse of A)
- ③ $r(A) = \mathbf{0}_{m \times n} \implies r = 0 \text{ or } A = \mathbf{0}_{m \times n}$

NOTES

- All above properties are very similar to the counterpart properties for real numbers

Properties of matrix multiplication

Theorem

Let A be $m \times n$ matrix, and let B and C have sizes for which the indicated sums and products are defined, then

- ① $A(BC) = (AB)C$, (*Associative law of multiplication*)
- ② $A(B + C) = AB + AC$, (*Left distributive property*),
- ③ $(A + B)C = AC + BC$, (*Right distributive property*),
- ④ $c(AB) = (cA)B = A(cB)$ for any scalar c , (*Associative property of scalar and matrix multiplication*)

NOTES

- For real numbers, the properties (2) and (3) are the same since the order for the multiplication of real numbers is irrelevant
- The real-number multiplication can satisfy above properties and there is a commutative property for the real-number multiplication, i.e., $cd = dc$

Properties of the identity matrix

Theorem

Let A be $m \times n$ matrix, then

- ① $AI_n = A$,
- ② $I_mA = A$,

NOTES

- The role of the real number 1 is similar to the identity matrix. However, 1 is unique in real numbers and there could be many identity matrices with different sizes.

Matrix Multiplication is Associative

Ex4: Calculate $(AB)C$ and $A(BC)$ for

$$A = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -2 & 1 \end{bmatrix}, \text{ and } C = \begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 4 \end{bmatrix}$$

Soln

$$\begin{aligned} (AB)C &= \left(\begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 3 & -2 & 1 \end{bmatrix} \right) \begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 4 \end{bmatrix} \\ &= \begin{bmatrix} -5 & 4 & 0 \\ -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 17 & 4 \\ 13 & 14 \end{bmatrix} \end{aligned}$$

Matrix Multiplication is Associative

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$$A = \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -2 & 1 \end{bmatrix}, \quad \text{and} \quad C = \begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 4 \end{bmatrix}$$

Soln

$$\begin{aligned} A(BC) &= \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \left(\begin{bmatrix} 1 & 0 & 2 \\ 3 & -2 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 4 \end{bmatrix} \right) \\ &= \begin{bmatrix} 1 & -2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ -7 & 2 \end{bmatrix} = \begin{bmatrix} 17 & 4 \\ 13 & 14 \end{bmatrix} \end{aligned}$$

Powers of a Matrix

Definition of A^k : repeated multiplication of a square matrix

If $A = [a_{ij}]_{n \times n}$ and if k is a positive integer then A^k denotes the product of k copies of A . That is,

$$A^k = \underbrace{A \dot{A} \cdots A}_k$$

Properties for A^k

① $A^j A^k = A^{j+k}$

② $(A^j)^k = A^{jk}$

where j and k are nonnegative integers and A^0 is assumed to be I .

Transpose of a matrix

Definition of Transpose of a matrix

If $A = [a_{ij}]_{m \times n}$, the **transpose** of A is the $n \times m$ matrix denoted by A^T , whose columns are formed from the corresponding rows of A . That is, if

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} \in M_{m \times n},$$

then

$$A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} & \cdots & a_{m1} \\ a_{12} & a_{22} & a_{32} & \cdots & a_{m2} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ a_{1n} & a_{2n} & a_{3n} & \cdots & a_{mn} \end{bmatrix} \in M_{n \times m},$$

Transpose of a matrix

Ex5: Calculate A^T , B^T and C^T

$$A = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 2 \\ 3 & -2 & 1 \\ 4 & 6 & 3 \end{bmatrix}, \text{ and } C = \begin{bmatrix} -1 & 0 \\ 3 & 1 \\ 2 & 4 \end{bmatrix}$$

Soln

①

$$A^T = [1 \quad 2]$$

②

$$B^T = \begin{bmatrix} 1 & 3 & 4 \\ 0 & -2 & 6 \\ 2 & 1 & 3 \end{bmatrix}$$

③

$$C^T = \begin{bmatrix} -1 & 3 & 2 \\ 0 & 1 & 4 \end{bmatrix}$$

Properties of transposes

Theorem

Let A be $m \times n$ matrix, then

- ① $(A^T)^T = A$,
- ② $(A + B)^T = A^T + B^T$,
- ③ $(cA)^T = cA^T$,
- ④ $(AB)^T = B^T A^T$

NOTES

- Properties (2) and (4) can be generalized to the sum or product of multiple matrices. For example, $(A + B + C)^T = A^T + B^T + C^T$ and $(ABC)^T = C^T B^T A^T$
- Since a real number also can be viewed as a 1×1 matrix, the transpose of a real number is itself, that is, for $a \in R$, $a^T = a$. In other words, transpose operation has actually no effect on real numbers

Transpose of a matrix

Ex6: Verify that $(AB)^T$ and $B^T A^T$ are equal

$$A = \begin{bmatrix} 2 & 1 & -2 \\ -1 & 0 & 3 \\ 0 & -2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 1 \\ 2 & -1 \\ 2 & 0 \end{bmatrix}$$

Soln

$$\begin{aligned} (AB)^T &= \left(\begin{bmatrix} 2 & 1 & -2 \\ -1 & 0 & 3 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & -1 \\ 2 & 0 \end{bmatrix} \right)^T \\ &= \begin{bmatrix} 2 & 1 \\ 6 & -1 \\ -1 & 2 \end{bmatrix}^T = \begin{bmatrix} 2 & 6 & -1 \\ 1 & -1 & 2 \end{bmatrix} \end{aligned}$$

Transpose of a matrix

Ex6: Verify that $(AB)^T$ and $B^T A^T$ are equal

$$A = \begin{bmatrix} 2 & 1 & -2 \\ -1 & 0 & 3 \\ 0 & -2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 1 \\ 2 & -1 \\ 2 & 0 \end{bmatrix}$$

Soln

$$\begin{aligned} B^T A^T &= \begin{bmatrix} 3 & 1 \\ 2 & -1 \\ 2 & 0 \end{bmatrix}^T \begin{bmatrix} 2 & 1 & -2 \\ -1 & 0 & 3 \\ 0 & -2 & 1 \end{bmatrix}^T \\ &= \begin{bmatrix} 3 & 2 & 2 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 & 0 \\ 1 & 0 & -2 \\ -2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 6 & -1 \\ 1 & -1 & 2 \end{bmatrix} \end{aligned}$$

Symmetric matrix and Skew-symmetric matrix

Symmetric matrix

A square matrix A is symmetric if $A = A^T$.

Skew-symmetric matrix

A square matrix A is skew-symmetric if $A^T = -A$.

Ex 7

If $A = \begin{bmatrix} 1 & 2 & 3 \\ a & 4 & 5 \\ b & c & 6 \end{bmatrix}$ is symmetric, find a, b, c ?

Symmetric matrix and Skew-symmetric matrix

Symmetric matrix

A square matrix A is symmetric if $A = A^T$.

Skew-symmetric matrix

A square matrix A is skew-symmetric if $A^T = -A$.

Ex 7

If $A = \begin{bmatrix} 1 & 2 & 3 \\ a & 4 & 5 \\ b & c & 6 \end{bmatrix}$ is symmetric, find a, b, c ?

Soln

$$A = \begin{bmatrix} 1 & 2 & 3 \\ a & 4 & 5 \\ b & c & 6 \end{bmatrix}, A^T = \begin{bmatrix} 1 & 2 & b \\ 2 & 4 & c \\ 3 & 5 & 6 \end{bmatrix}, A = A^T \implies a = 2, b = 3, c = 5.$$

Symmetric matrix and Skew-symmetric matrix

Ex 8

If $A = \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ b & c & 0 \end{bmatrix}$ is a skew-symmetric, find a, b, c ?

Symmetric matrix and Skew-symmetric matrix

Ex 8

If $A = \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ b & c & 0 \end{bmatrix}$ is a skew-symmetric, find a, b, c ?

Soln

$$A = \begin{bmatrix} 0 & 1 & 2 \\ a & 0 & 3 \\ b & c & 0 \end{bmatrix}, -A^T = \begin{bmatrix} 0 & -a & -b \\ -1 & 0 & -c \\ -2 & -3 & 0 \end{bmatrix}, A = -A^T \implies a = -1, b = -2, c = -3.$$

Orthogonal matrix

Definition of orthogonal

A matrix A is called orthogonal if $AA^T = A^T A = I$, i.e., $A^T = A^{-1}$.

NOTES

- The matrix A could be with any size, i.e., it is not necessary for A to be a square matrix.
- The matrix A could be with any size, i.e., it is not necessary for A to be a square matrix.
- AA^T must be symmetric
- In fact, AA^T must be a square matrix.

Pf:

$$(AA^T)^T = (A^T)^T A^T = AA^T$$

$\therefore AA^T$ is symmetric

Remarks

- Before finishing this section, two properties will be discussed, which are held for real numbers, but not for matrices: the first is the commutative property of matrix multiplication and the second is the cancellation law.
- Real number:
 $ab = ba$ (Commutative property of real-number multiplication)

- Matrix:

$$\underbrace{AB}_{m \times n \ n \times p} \neq \underbrace{BA}_{n \times p \ m \times n}$$

Three results for BA

- If $m \neq p$, then AB is defined but BA is undefined
- If $m = p, m \neq n$, then $AB \in M_{m \times m}$, $BA \in M_{n \times n}$, (Sizes are not the same)
- If $m = p = n, m \neq n$, then $AB \in M_{m \times m}$, $BA \in M_{m \times m}$ (Sizes are the same, but resulting matrices may not be equal)

Ex 9

Show that AB and BA are not equal for the following matrices.

$$A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix}$$

Ex 9

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Soln

$$AB = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 4 & -4 \end{bmatrix}$$

$$BA = \begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 7 \\ 4 & -2 \end{bmatrix}$$

Hence $AB \neq BA$ (noncommutativity of matrix multiplication)

- $A + B = B + A$
(the commutative law of matrix addition)
- $AB \neq BA$
(the matrix multiplication is not with the commutative law) (so the order of matrix multiplication is very important)

NOTE

- This property is different from the property for the multiplication operations of real numbers, for which the order of multiplication makes no difference

Cancellation law

- Real number:

$$ac = bc \text{ and } c \neq 0 \implies a = b)$$

(Cancellation law for real numbers)

- Matrix:

$$AC = BC \text{ and } C \neq 0 \text{ (C is not a zero matrix)}$$

- ① If C is invertible, then $A = B$,
- ② If C is not invertible, then $A \neq B$,
(Cancellation law is not necessary to be valid)

NOTE

- We shall more about invertible in the next lecture.

Cancellation law

Ex 10: An example in which cancellation is not valid
Show that $AC = BC$.

$$A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix} \quad C = \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix}$$

Cancellation law

Ex 10: An example in which cancellation is not valid
Show that $AC = BC$.

$$A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix} \quad C = \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix}$$

Soln

$$AC = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ -1 & 2 \end{bmatrix}$$

$$BC = \begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ -1 & 2 \end{bmatrix}$$

So, although $AC = BC, A \neq B$

Exercise 1

Exercise 1

- ① Since vectors in R^n may be regarded as $n \times 1$ matrices, the properties of transposes apply to vectors, too. let

$$A = \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} \quad \text{and} \quad \mathbf{x} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

Compute $(A\mathbf{x})^T$, $\mathbf{x}\mathbf{x}^T A^T$, $\mathbf{x}\mathbf{x}^T$ and $\mathbf{x}^T \mathbf{x}$. Is $A^T \mathbf{x}^T$ defined?

- ② Let A be a 4×4 matrix and let $\mathbf{x}$ be a vector in R^4 . What is the faster way to compute $A^2 \mathbf{x}$? Count the number of multiplication.
- ③ If $A = \begin{bmatrix} 1 & -3 \\ -3 & 5 \end{bmatrix}$ and $AB = \begin{bmatrix} -3 & -11 \\ 1 & 7 \end{bmatrix}$, determine the first and second columns of B .

Inverse Matrix

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$$AB = I \text{ and } BA = I,$$

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- B is uniquely determined by A . Inverse is Unique. The inverse of A is denoted by A^{-1} ,
- *Singular matrix* – not invertible,
- *Nonsingular matrix* – invertible.

Definition

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NOTE

- Definition of the inverse of a matrix is similar to that of the inverse of a scalar, $(c(1/c))$.
- Since there is no inverse (or said multiplicative inverse) for the real number 0, you can “imagine” that noninvertible matrices act a similar role to the real number 0 in some

Inverse Matrix

Ex 11: Verify AB and BA if

$$A = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix} \text{ and } B = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix}.$$

Inverse Matrix

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$$AB = \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix} \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Thus, $B = A^{-1}$.

The Inverse of a Matrix is Unique

Theorem

If B and C are both inverses of the matrix A , then $B = C$.

Proof.

$$AB = I$$

$$C(AB) = C(I)$$

$(CA)B = C$, (associative property of matrix multiplication and the property for the identity matrix)

$$IB = C$$

$$B = C$$

Consequently, the inverse of a matrix is unique. □

Inverse of a 2×2 Matrix

Theorem

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$ (determine), then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

If $ad - bc = 0$, then A is not invertible.

Ex12: Find the inverse of $A = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$

Properties of inverse matrices

Theorem

If A is an invertible matrix, k is a positive integer, and c is a scalar, then

- ① A^{-1} is invertible and $(A^{-1})^{-1} = A$.
- ② A^k is invertible and $(A^k)^{-1} = (A^{-1})^k$.
- ③ cA is invertible given $c \neq 0$ and $(cA)^{-1} = \frac{1}{c}A^{-1}$.
- ④ A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.

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- 2 A^k is invertible and $(A^k)^{-1} = (A^{-1})^k$.
- 3 cA is invertible given $c \neq 0$ and $(cA)^{-1} = \frac{1}{c}A^{-1}$.
- 4 A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.

- 1 $A^{-1}A = I$
- 2 $A^k(A^{-1})^k = I$
- 3 $cA\frac{1}{c}A^{-1} = I$
- 4 $A^T(A^{-1})^T = I$

The inverse of a product

Theorem

If A and B are invertible matrices of order n , then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$

The inverse of a product

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Proof.

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = A(I)A^{-1} = (AI)A^{-1} = AA^{-1} = I$$



The inverse of a product

Theorem

If A and B are invertible matrices of order n , then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$

Proof.

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NOTE

- It can be generalized to the product of multiple matrices

$$(A_1A_2 \cdots A_n)^{-1} = A_n^{-1} \cdots A_2^{-1}A_1^{-1}$$

- It is similar to the results of the transpose of the products of multiple matrices

$$(A_1A_2 \cdots A_n)^T = A_n^T \cdots A_2^T A_1^T$$

Cancellation properties for matrix multiplication

Theorem

If C is an invertible matrix, then the following properties hold:

- ① *If $AC = BC$, then $A = B$ (right cancellation property)*
- ② *If $CA = CB$, then $A = B$ (left cancellation property)*

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Proof.

$$AC = BC$$

$$(AC)C^{-1} = (BC)C^{-1}, \text{ (} C \text{ is invertible, so } C^{-1} \text{ exists)}$$

$$A(CC^{-1}) = B(CC^{-1}), \text{ (Associative property of matrix multiplication),}$$

$$AI = BI,$$

$$A = B.$$

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$$A(CC^{-1}) = B(CC^{-1}), \text{ (Associative property of matrix multiplication),}$$

$$AI = BI,$$

$$A = B.$$

Cancellation properties for matrix multiplication

NOTE

- If C is not invertible, then cancellation is not valid.

Systems of equations with a unique solution

Theorem

If A is an invertible matrix, then the system of linear equations $A\mathbf{x} = \mathbf{b}$, $\mathbf{b} \in R^n$ has a unique solution given by $\mathbf{x} = A^{-1}\mathbf{b}$.

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Proof.

$$\begin{aligned}A\mathbf{x} &= \mathbf{b}, \\A^{-1}A\mathbf{x} &= A^{-1}\mathbf{b}, \quad (A \text{ is nonsingular}) \\I\mathbf{x} &= A^{-1}\mathbf{b}, \\ \mathbf{x} &= A^{-1}\mathbf{b}.\end{aligned}$$

If $\mathbf{x}_1$ and $\mathbf{x}_2$ were two solutions to the linear equation $A\mathbf{x} = \mathbf{b}$, then

$$A\mathbf{x}_1 = \mathbf{b} = A\mathbf{x}_2 \implies \mathbf{x}_1 = \mathbf{x}_2, \text{ (left cancellation property)}$$

Definition

Elementary matrix

An $n \times n$ matrix is called an elementary matrix if it can be obtained from the identity matrix I by a single elementary row operation.

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Three row elementary matrices:

- $E_{i,j} = I_{i,j}(I)$, (interchange two rows).
- $E_i^{(k)} = M_i^{(k)}(I)$, ($k \neq 0$), (multiply a row by a nonzero constant).
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- $E_{i,j}^{(k)} = A_{i,j}^{(k)}(I)$, (add a multiple of a row to another row).

NOTE

- Perform only a single elementary row operation on the identity matrix.
- Since the identity matrix is a square matrix, elementary matrices must be square matrices.

Elementary Matrices

Ex 12: Elementary matrices and nonelementary matrices

$$\textcircled{1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Elementary Matrices

Ex 12: Elementary matrices and nonelementary matrices

① $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ Yes ($M_2^{(3)}(I_3)$)

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③ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ No (Row multiplication must be with a nonzero constant)

④ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

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Ex 12: Elementary matrices and nonelementary matrices

① $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ Yes ($M_2^{(3)}(I_3)$)

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④ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ Yes ($I_{2,3}(I_3)$)

Elementary Matrices

Ex 12 (Conti.): Elementary matrices and nonelementary matrices

① $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$

Elementary Matrices

Ex 12 (Conti.): Elementary matrices and nonelementary matrices

① $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ Yes ($A_{1,2}^{(2)}(I_2)$)

② $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

Elementary Matrices

Ex 12 (Conti.): Elementary matrices and nonelementary matrices

① $\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$ Yes ($A_{1,2}^{(2)}(I_2)$)

② $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ No (Use two elementary row operations, $M_2^{(2)}$ and $M_3^{(-1)}$).

NOTE

- Elementary matrices are useful because they enable you to use matrix multiplication to perform elementary row operations.

Representing elementary row operations

Theorem

Let E be the elementary matrix obtained by performing an elementary row operation on I_m . If that same elementary row operation is performed on an $m \times n$ matrix A , then the resulting matrix equals the product EA .

Note

- The elementary matrix must be multiplied to the left side of the matrix A

Using elementary matrices

Ex 13: Find a sequence of elementary matrices that can be used to rewrite the matrix A in its row-echelon form

$$A = \begin{bmatrix} 0 & 1 & 3 & 5 \\ 1 & -3 & 0 & 2 \\ 2 & -6 & 2 & 0 \end{bmatrix} \xrightarrow{I_{1,2}, A_{1,3}^{(-2)}, M_3^{(1/2)}} \begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

Soln

$$E_1 = I_{1,2}(I_3) = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, E_2 = A_{1,3}^{(-2)}(I_3) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}, E_3 = M_3^{(1/2)}(I_3) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/2 \end{bmatrix}$$

Using elementary matrices

$$A_1 = I_{1,2}(A) = E_1 A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 3 & 5 \\ 1 & -3 & 0 & 2 \\ 2 & -6 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 2 & -6 & 2 & 0 \end{bmatrix}$$

$$A_2 = A_{1,3}^{(-2)}(A_1) = E_2 A_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 2 & -6 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

$$A_3 = M_3^{(1/2)}(A_2) = E_3 A_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1/2 \end{bmatrix} \begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 1 & 3 & 5 \\ 0 & 0 & 1 & -3 \end{bmatrix} = B$$

$$\therefore B = E_3 E_2 E_1 A \text{ or } B = M_3^{(1/2)}(A_{1,3}^{(-2)}(I_{1,2}(A)))$$

NOTE

- This example demonstrates that we can obtain the same effect of performing elementary row operations by multiplying corresponding elementary matrices to the left side of the target matrix A .

Representing elementary row operations

Row-equivalent

Matrix B is row-equivalent to A if there exists a finite number of elementary matrices such that

$$B = E_k E_{k-1} \cdots E_2 E_1$$

Representing elementary row operations

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Matrix B is row-equivalent to A if there exists a finite number of elementary matrices such that

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Note

- This row-equivalent property is from the row-equivalent property after performing a finite sequence of the elementary row operations

Theorem

If A is an $n \times n$ matrix, then the following statements are equivalent

- ① A is invertible
- ② $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ column vector $\mathbf{b}$
- ③ $A\mathbf{x} = \mathbf{b}$ has only the trivial solution
- ④ A is row-equivalent to I_n
- ⑤ A can be written as the product of elementary matrices

An Algorithm for Finding A^{-1}

Theorem

Row reduce the augmented matrix $[A \ I]$. If A is row equivalent to I , then $[A \ I]$ is row equivalent to $[I \ A^{-1}]$. Otherwise, A does not have an inverse.

Exercise 2

Exercise 2

- ① Use the determinants to determine which of the following matrices are invertible.

$$A = \begin{bmatrix} 3 & -9 \\ 2 & 6 \end{bmatrix} \quad B = \begin{bmatrix} 4 & -9 \\ 5 & 5 \end{bmatrix} \quad C = \begin{bmatrix} 6 & -9 \\ -4 & 6 \end{bmatrix}$$

- ② Find the inverse of the matrix $A = \begin{bmatrix} 1 & -2 & -1 \\ -1 & 5 & 6 \\ 5 & -4 & 5 \end{bmatrix}$, if it exist.

- ③ Suppose B, C are invertible, solve X in terms of A, B, C, D

① $BXC = A$

② $B(A + X)C = D$